48V Electrical Architecture

As-of date: `2026-03-20`

Purpose: hold the finalized, concise `48V` house and alternator architecture in one place so wiring, protection, shutdown behavior, and BOM references are easy to understand without re-reading the historical trade studies.

Related docs:

- `docs/implementation/ELECTRICAL_overview_diagram.md`
- `docs/implementation/ELECTRICAL_fuse_schedule.md`
- `docs/core/SYSTEMS.md`
- `docs/core/TRACKING.md`
- `bom/bom_estimated_items.csv`

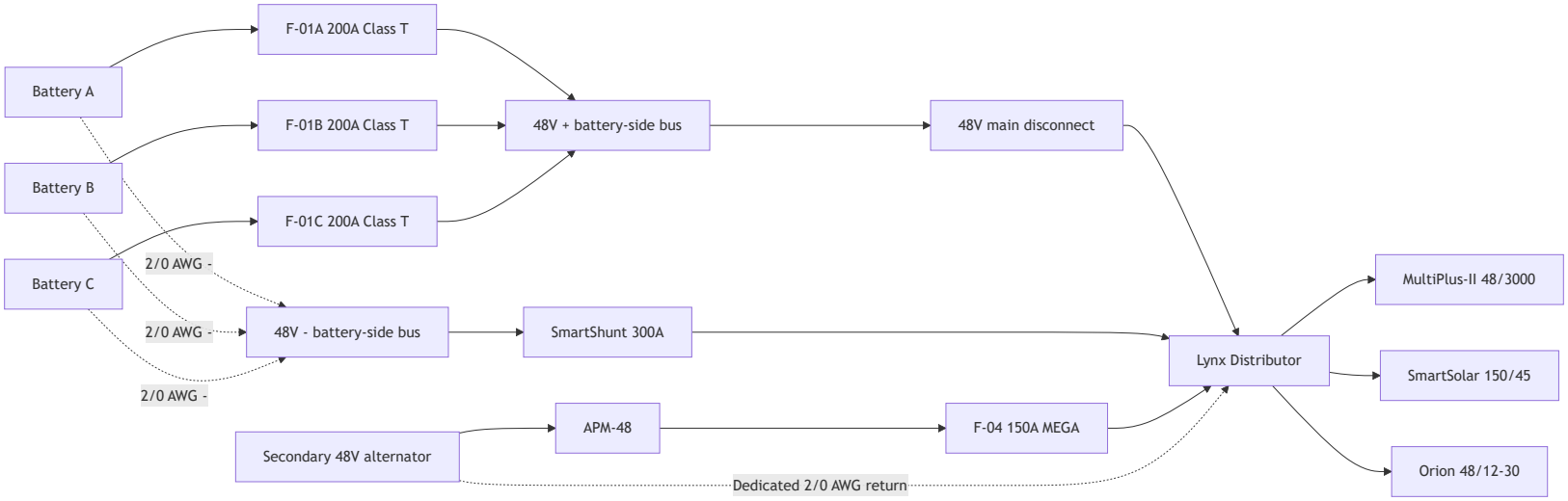
Final direction

- House architecture stays `48V` core with `3x 51.2V 100Ah` batteries in parallel.
- Alternator charging is the dedicated `48V` secondary alternator path, not Sterling B2B.
- Protection baseline is `Mechman + WS500 + Balmar APM-48`.
- Manual alternator-charge enable/disable is through Ford `Upfitter Switch #3`.
- `Upfitter #3` is a low-current control signal only. It does not carry alternator output current.
- `WS500` white `Feature-In` is reserved for future fault-interlock work, not required in Phase 1.

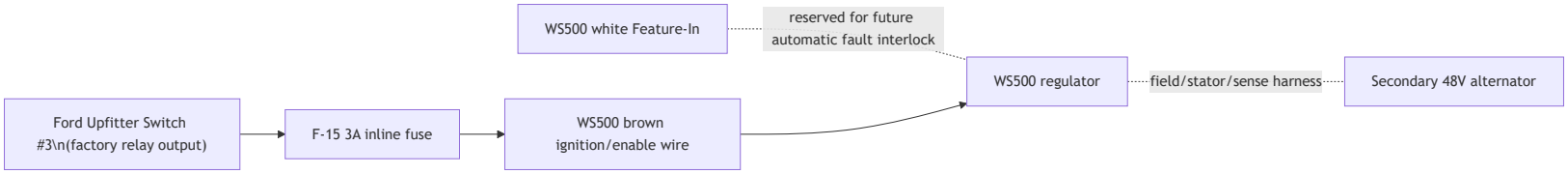
Locked component set

Function	Locked baseline	BOM row(s)
House batteries	`3x` Dumfume `51.2V 100Ah` LiFePO4	`3`
Main house disconnect	Victron `275A` battery switch	`5`
Main fused distribution	Victron Lynx Distributor `M10`	`6`
Battery branch protection	`F-01A/B/C 200A` Class T (provisional pending final battery limit confirmation)	`7, 105`
Inverter/charger	MultiPlus-II `48/3000/35-50`	`12`
48V to 12V charger	Orion-Tr Smart `48/12-30`	`20`
Monitoring	Cerbo GX + SmartShunt `300A`	`22, 23`
Alternator kit	Mechman `48V` secondary alternator kit with `WS500`	`168`
Load-dump clamp	Balmar `APM-48`	`169`
Alternator branch fuse	`F-04 150A` MEGA (`58V/80V`) in Lynx Slot 3	`170`
WS500 low-current fuse set	`F-12, F-13, F-14`	`171`
WS500 Upfitter #3 control kit	`F-15` + control wire + holder/terminals	`176`

48V power path



Alternator control path



Control logic

- `Upfitter #3 ON` : `WS500` brown wire is energized, regulator is allowed to run, alternator charging can occur.
- `Upfitter #3 OFF` : `WS500` is disabled through the brown wire, regulator field output collapses, and alternator charging stops.
- Use `Upfitter #3` as the manual alternator-charge shutdown control.
- Do not use the main `48V` disconnect as the first alternator shutdown method while the engine is running and the secondary alternator is charging.

Why `Upfitter #3`

- It is already a factory-switched, relay-driven `12V` control source.
- It keeps the WS500 control in the truck cab without adding a separate aftermarket switch.
- It is suitable for a low-current enable signal, but not for any high-current alternator or battery conductor.
- A local inline fuse is still required because the factory upfitter circuit fuse is much larger than the small-gauge WS500 control wire.

Fusing and wire baseline

ID	Function	Locked value	Wire basis
`F-01A/B/C`	Battery branch positive protection	`200A` Class T provisional	`2/0 AWG`
`F-04`	Alternator branch into Lynx Slot 3	`150A` MEGA (`58V/80V`)	`2/0 AWG`
`F-12`	WS500 power lead	`10A` baseline (`15A` if required by alternator case)	harness lead
`F-13`	WS500 battery positive sense	`3A`	harness lead
`F-14`	WS500 current-sense lead	`5A` where required by selected shunt layout	harness lead
`F-15`	Upfitter #3 to WS500 brown ignition wire	`3A` inline ATO/ATC	`16 AWG` TXL/GXL control wire

Major conductors

- Battery branch and main `48V` trunk wiring: `2/0 AWG` .
- Secondary alternator positive path (`ALT B+ -> APM-48 -> F-04 -> Lynx`): `2/0 AWG` , `~20 ft` one-way planning basis.
- Secondary alternator dedicated negative return (`ALT B- -> Lynx -`): `2/0 AWG` , `~20 ft` one-way planning basis.
- Orion `48V` feeder and MPPT battery leads: `6 AWG` .

- Upfitter #3 control lead to WS500 brown wire: `16 AWG` TXL/GXL planning basis, `~6 ft` one-way assumed until measured.

APM-48 wiring intent

- Mount the `APM-48` at the alternator end of the `48V` branch, as close to the alternator output as practical.
- Connect APM red to alternator `B+`.
- Connect APM black to alternator `B-`, or to the approved alternator ground/case point if the alternator is not isolated-ground.
- Do not stack the APM ring terminals under the main battery cable lugs unless the product instructions for the exact unit explicitly allow it.

Normal operating sequence

1. Main `48V` disconnect closed.
2. `Upfitter #3` switched `ON`.
3. Engine running.
4. `WS500` enabled and regulating.
5. Alternator charges the house bank through `APM-48` and `F-04`.

Fault and shutdown sequence

If a battery pack trips or an alternator fault is suspected:

1. Switch `Upfitter #3` `OFF` to disable the `WS500`.
2. Wait for alternator charge current to collapse.
3. Open the main `48V` disconnect only after alternator charging is no longer active, if full house shutdown is required.

Reason:

- The regulator should be shut down first.
- The `APM-48` is a protection layer, not the primary shutdown method.

One-battery-trip behavior

- If one battery BMS disconnects but the other two remain online, the `48V` bus should stay up.
- The system does not immediately become a full-bank load dump event just because one battery disappears.
- The practical effect is that charge and discharge current redistribute to the remaining batteries.
- The real hazard is a cascade where the second and third battery also disconnect under active alternator charge.

What is still not fully closed

- Final Mechman kit fitment/content confirmation for the exact truck.
- Final `WS500` harness polarity confirmation (`PH` vs `NH`).
- Official vendor confirmation that the documented Dumfume battery/BMS behavior is acceptable with the `WS500`.
- Exact alternator negative/case isolation behavior in the installed Mechman kit.
- Whether future automatic fault-interlock logic should be added on the WS500 `Feature-In` or through an external relay.

Rule for future edits

- Update this file first for any `48V` architecture, alternator-control, or shutdown-strategy change.
- Keep the implementation files for wiring detail and fuse placement, not for high-level architecture decisions.